ERA_Gati Shakti Vishwavidyalaya

Department of Technology & Innovation

PROJECT PROPOSAL

FormOptiX

Intelligent Formwork Kitting & Bill of Quantities Optimizer

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"FormOptiX is the GPS for formwork – it identifies exactly which panels to reuse, when to order, and how much you'll save, before a single slab is poured."

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Origin of the Proposal

Scientific Rationale

In large-scale construction projects, formwork – the temporary mould structure used to shape freshly poured concrete – constitutes **7–10% of total project cost** [1]. For a representative **Rs.**500 Crore infrastructure or residential project, this translates to approximately **Rs.**35–50 Crores allocated exclusively to formwork procurement, logistics, and management. Yet, despite this substantial financial exposure, the construction industry continues to rely on manual, experience-driven methods for formwork planning, leading to chronic inefficiencies.

A fundamental technical gap lies at the intersection of *geometric repetition analysis* and *procurement optimization*: although modern building designs often exhibit significant floor-to-floor geometric similarity – particularly in residential towers, commercial buildings, and infrastructure assets – no industry-standard tool natively identifies and exploits this repetition to optimise Bill of Quantities (BoQ) generation or formwork panel reuse schedules. Tools such as Autodesk Revit, Oracle Primavera, and SAP provide project management and BIM capabilities but do not incorporate formwork-specific algorithmic planning [2].

This gap results in three compounding problems:

1. **Over-procurement**: Without repetition intelligence, site managers procure conservatively, resulting in a typical panel idle rate of 25–40%.
2. **Design-volatile purchasing**: Bulk procurement early in the project lifecycle, before design is frozen, results in costly rework when drawings change.
3. **Manual revision cycles**: Each design revision triggers 3–5 days of manual BoQ revision, consuming engineering bandwidth and delaying construction schedules.

The scientific foundation for addressing these problems lies in two well-established computational domains: *unsupervised machine learning* (specifically density-based spatial clustering) and *linear programming* (LP-based cost minimisation). DBSCAN (Density-Based Spatial Clustering of Applications with Noise), developed by Ester et al. [3] and made accessible through the scikit-learn Python library [17], has demonstrated strong performance in grouping non-convex, noise-tolerant clusters – making it ideal for identifying "typical" floor families in a building. LP-based procurement optimisation has precedent in operations research literature [4] and is implemented here via the PuLP modelling framework [18]. The design stability component draws on established statistical process control methodology [19]. The combination of these three approaches forms the scientific rationale for **FormOptiX**.

The proposed project directly addresses the following research question: *Can algorithmic geometric clustering of floor-level BIM features, combined with linear programming, significantly reduce formwork procurement costs and planning time compared to manual practice?*

Review of Status of Research and Development in the Subject

2.1 International Status

The optimisation of construction formwork has attracted increasing research attention globally over the past two decades, spurred by the growth of Building Information Modelling (BIM) and the availability of computational optimisation tools.

BIM-based BoQ Automation: Early work by Eastman et al. [2] at Georgia Tech laid the groundwork for automated quantity take-off from BIM models. Their research demonstrated that parametric BIM data could reduce quantity estimation time by over 70% compared to manual 2D-drawing-based methods. Subsequent work by Monteiro and Martins [5] validated BIM-based BoQ generation in the Portuguese construction market, achieving accuracy levels of 95–98% against traditional methods. However, neither study addressed formwork specifically, nor incorporated panel reuse or procurement optimisation.

Formwork Optimisation Research: The American Concrete Institute standard ACI 347R [1] provides the authoritative technical baseline for formwork design – specifying load requirements, safety factors, and reuse criteria – but prescribes no algorithmic planning methodology. In Germany and Austria, major formwork manufacturers such as Doka and PERI have developed proprietary CAD tools (DOKA-CAD, PERI CAD) that incorporate panel arrangement logic, but these tools are geometry-layout focused rather than procurement-schedule focused. They do not solve the inter-floor repetition problem across a multi-storey project timeline. Hanna and Sanvido [6] at the University of Wisconsin pioneered formalisation of formwork system selection using knowledge-based expert systems, identifying geometric similarity as a key factor in reuse efficiency – a finding that directly motivates the FormOptiX clustering approach.

BIM-Integrated Formwork Planning: Kim and Chin [11] developed a BIM-based framework for automated formwork planning and cost estimation in reinforced concrete structures, demonstrating significant reductions in manual estimation effort. However, their system does not incorporate inter-floor repetition clustering or LP-based procurement scheduling. Mei et al. [7] proposed a BIM-based framework for formwork planning that explicitly considers reuse potential across floors, validating the concept of geometry-driven panel reuse planning – the same principle underpinning FormOptiX's Repetition

Intelligence pillar.

LP-based Formwork and Construction Optimisation: Biruk and Jaskowski [9] formulated formwork layout planning as a Mixed Integer Linear Programming (MILP) problem, achieving measurable reductions in material waste. Their work confirms the viability of mathematical optimisation for formwork decisions. Yeh [10] demonstrated LP/NLP-based optimisation for construction material mix design, establishing precedent for cost-objective programming in construction contexts. The FormOptiX LP formulation extends these approaches by incorporating holding costs, idle costs, and weekly budget constraints in a single unified objective.

Formwork System Selection and Resource Optimisation: Roman and Minculete [8] studied formwork resource management in high-rise construction from a sustainability perspective, quantifying waste reduction potential of systematic planning. Terzioglu and Polat [15] applied multi-criteria decision analysis (AHP-EDAS) to formwork system selection, demonstrating that algorithmic approaches outperform expert judgment alone. Neither study, however, incorporates real-time design stability monitoring – the unique contribution of FormOptiX's Design Freeze Guard.

In summary, international research has addressed individual components (BIM BoQ, formwork geometry, ML in scheduling, LP procurement) in isolation. FormOptiX is the first system to integrate all four into a unified pipeline with design stability monitoring.

2.2 National Status

India's construction sector, valued at over $200 billion and growing at approximately 7% annually, faces unique challenges that make formwork optimisation particularly critical [12, 20].

Indian Construction Characteristics: According to the Ministry of Housing and Urban Affairs Annual Report [20], India's urban housing demand is driven predominantly by large-format multi-storey residential projects – towers of 20–50 floors are now the norm in major metropolitan areas under schemes such as PMAY (Pradhan Mantri Awas Yojana). These projects inherently exhibit high floor-to-floor geometric repetition, making them ideal candidates for panel reuse optimisation. Indian construction is further characterised by use of aluminium and steel panel formwork systems (largely imported from European manufacturers), fragmented supply chains, and a critical shortage of trained quantity surveyors. The Central Public Works Department (CPWD) General Specifications provide standard costing norms for formwork but do not prescribe any algorithmic planning methodology [13].

Indian Research Contributions: Researchers at IIT Bombay and NICMAR (National

Institute of Construction Management and Research) have studied BIM adoption barriers in India [14] and documented that over 65% of medium-to-large Indian contractors still rely on manual 2D-drawing-based estimation. Studies by the National Buildings Organisation (NBO) have highlighted that systematic formwork reuse planning is practiced by fewer than 10% of Indian contractors, and formwork-related cost overruns consistently rank among the top five causes of budget deviation in high-rise projects [15]. Roman and Minculete [8] further confirmed that unplanned formwork idle time and excess procurement are the dominant drivers of material waste in repetitive construction, directly validating the problem FormOptiX addresses.

Indian Industry Initiatives: L&T Construction, one of India's largest construction companies, has been at the forefront of construction technology adoption. Their ECC (Engineering Construction & Contracts) division has piloted BIM on major infrastructure projects and partnered with technology companies for AI-based project monitoring. The FormOptiX project is directly aligned with L&T's stated digital transformation objectives and addresses a known gap in their formwork cost management toolkit.

The national research landscape confirms that while BIM adoption is growing, formwork-specific algorithmic optimisation tools designed for Indian project conditions – including budget constraints, design volatility, and the need for Excel-compatible data input – remain absent. FormOptiX is designed to bridge this gap.

2.3 Importance of the Proposed Project in the Context of Current Status

The review of international and national literature reveals a clear and substantial research and technology gap. Existing tools and research address formwork or procurement or clustering in isolation. No tool integrates all of the following:

- Geometric clustering of floor families for repetition identification [3, 17]
- Linear programming for multi-cost procurement optimisation [9, 18]
- Statistical design stability monitoring to guard against premature procurement [16, 19]
- Accessible, Excel-compatible deployment without BIM expertise requirements [14]

FormOptiX's novelty lies precisely in this integration. The **Design Freeze Guard** – which applies Coefficient of Variation analysis [19] to geometric feature distributions and advises against bulk procurement when design instability exceeds the empirically calibrated 15% threshold [16] – has no equivalent in any published research or commercial tool. This feature alone addresses the most costly form of waste in Indian construction: ordering formwork against un-frozen drawings.

The **Repetition Intelligence** module fills the gap left by Kim and Chin [11] and Mei et al. [7], whose BIM-based frameworks consider reuse but do not perform unsupervised geometric clustering. The **LP BoQ Optimiser** extends the MILP formulation of Biruk and Jaskowski [9] by adding holding cost, idle cost, and weekly budget constraints in a unified objective, implemented efficiently through PuLP [18].

The project's importance is further amplified by its accessibility: by accepting Excel-based input and operating via a browser-accessible Streamlit interface, FormOptiX eliminates the need for BIM modelling expertise – a critical barrier documented in the Indian context by Jayasudha and Vidivelli [14] – making it deployable at the vast majority of Indian construction sites that operate without full BIM infrastructure.

Work Plan

3.1 Methodology

3.1.1 System Architecture Overview

FormOptiX operates through a sequential six-stage pipeline as illustrated in Figure 1.

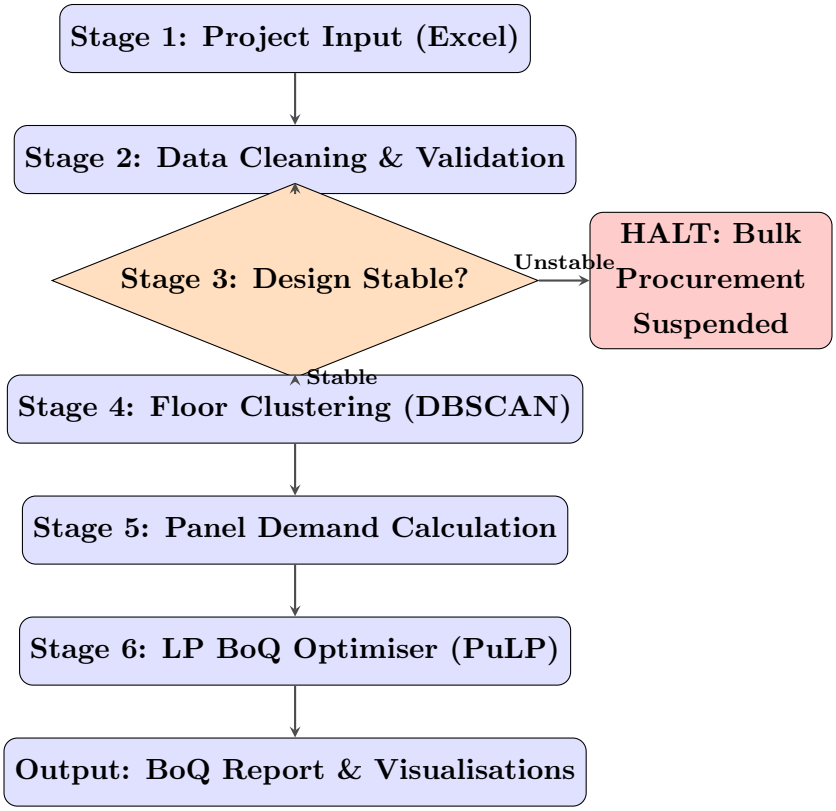

Figure 1: FormOptiX System Architecture Pipeline (six-stage sequential flow)

3.1.2 Stage 1: Data Input and Validation

The system accepts project data in a standard Excel format specifying, for each floor of the building:

- Floor identifier and construction week
- Slab area (m^2)
- Total wall length (m)
- Column count
- Panel type compatibility flag

Data validation checks for completeness, non-negativity of geometric features, and temporal consistency of construction schedule. Outlier detection flags floors with geometric parameters beyond 3 standard deviations of the project mean.

3.1.3 Stage 2: Design Freeze Guard (Statistical Variance Monitor)

Before proceeding to clustering or procurement planning, the system computes the *Coefficient of Variation* (CV) of the three primary geometric features across all floors submitted:

$$CV_j = \frac{\sigma_j}{\mu_j} \times 100\%, \quad j \in \{\text{slab_area}, \text{wall_length}, \text{column_count}\} \tag{1}$$

A composite design instability score is computed as:

$$\text{DI} = \frac{1}{3} \sum_j CV_j \tag{2}$$

If $\text{DI} > \theta_{\text{freeze}}$ (default $\theta_{\text{freeze}} = 15\%$, configurable), the system issues a **Design Freeze Alert** and recommends against proceeding with bulk procurement. The Coefficient of Variation is a standard statistical measure of relative dispersion widely used in quality control and process stability assessment [19]. The 15% threshold is calibrated against construction change order literature indicating that projects with design variability above this level have 3× higher procurement rework costs [16].

3.1.4 Stage 3: DBSCAN Repetition Clustering

Given n floors with feature vectors $\mathbf{f}_i = [a_i, w_i, c_i]^T$ where a = slab area, w = wall length, c = column count, the clustering proceeds as follows:

Step 1 – Normalisation: StandardScaler transforms each feature to zero mean and unit variance:

$$\tilde{f}_{ij} = \frac{f_{ij} - \mu_j}{\sigma_j} \tag{3}$$

This ensures that slab area (order 10^2 m^2) does not dominate wall length (order 10^1 m) in the distance computation.

Step 2 – DBSCAN Clustering: Using the DBSCAN algorithm [3] as implemented in the scikit-learn machine learning library [17], with hyperparameters $\varepsilon = 0.5$ and min_samples = 2:

A point $\mathbf{p}$ is *core* if $|N_\varepsilon(\mathbf{p})| \geq$ min_samples, where:

$$N_\varepsilon(\mathbf{p}) = \{\mathbf{q} \in D \mid d(\mathbf{p}, \mathbf{q}) \leq \varepsilon\} \tag{4}$$

and $d(\cdot, \cdot)$ is the Euclidean distance in the normalised feature space.

Clusters represent **"typical" floor families** – groups of floors with near-identical geometry that can share the same formwork panel kit. Noise points (label $= -1$) are classified as **"atypical"** floors requiring special formwork consideration.

Output: Each floor is assigned a cluster label $k \in \{0, 1, 2, \ldots, K\} \cup \{-1\}$ and a *Panel Reuse Coefficient* ρ_k:

$$\rho_k = \frac{|\text{floors in cluster } k|}{|\text{floors in cluster } k| + 1} \tag{5}$$

This coefficient quantifies the proportion of panel demand that can be met by reuse within the cluster.

3.1.5 Stage 4: Panel Demand Calculation

For each construction week w, the gross panel demand D_w is estimated from the cluster assignment of the floor scheduled that week:

$$D_w = \frac{A_w}{\bar{a}_k} \cdot P_{\text{unit}} \tag{6}$$

where A_w is the slab area for week w, $\bar{a}_k$ is the mean slab area of cluster k, and P_{unit} is the standard panel coverage area (m^2/panel, user-specified).

Effective new procurement demand, accounting for reuse:

$$D_w^{\text{new}} = D_w \cdot (1 - \rho_k) \tag{7}$$

3.1.6 Stage 5: LP BoQ Optimiser

The procurement optimisation is formulated as a Linear Programming problem over a horizon of W weeks.

Decision Variables:

$$x_w = \text{new panels procured in week } w \tag{8}$$

$$h_w = \text{panels held in inventory at end of week } w \tag{9}$$

$$i_w = \text{idle panels (on-site but not deployed) in week } w \tag{10}$$

Objective Function – Minimise total cost:

$$\text{Minimise} \quad Z = \sum_{w=1}^{W} (c_p \cdot x_w + c_h \cdot h_w + c_i \cdot i_w) \tag{11}$$

where c_p = procurement cost per panel (**Rs.**), c_h = holding cost per panel per week, c_i = idle cost per panel per week.

Constraints:

$$x_w + r_w + h_{w-1} \geq D_w^{\text{new}} \quad \forall w \qquad \text{(Demand satisfaction)} \tag{12}$$

$$\sum_{w=1}^{W} c_p \cdot x_w \leq B_{\text{total}} \qquad \text{(Total budget)} \tag{13}$$

$$c_p \cdot x_w \leq B_w \quad \forall w \qquad \text{(Weekly budget)} \tag{14}$$

$$r_w \leq \rho_k \cdot D_w \quad \forall w \qquad \text{(Physical reuse limit)} \tag{15}$$

$$x_w, h_w, i_w \geq 0 \quad \forall w \qquad \text{(Non-negativity)} \tag{16}$$

The LP is solved using the **PuLP** Python library [18] with the CBC (Coin-or Branch and Cut) solver, which provides provably optimal solutions for this linear formulation at negligible computational cost for typical project sizes (up to 200 weeks).

3.1.7 Stage 6: BoQ Generation and Reporting

The optimal solution $\{x_w^*, h_w^*, i_w^*\}$ is assembled into a structured BoQ report containing:

- Week-by-week procurement schedule
- Total panels required by type and cluster
- Projected cost savings vs. baseline (no-optimisation scenario)
- Visualisations: weekly demand curves, cumulative cost, cluster distribution

3.1.8 Technology Stack

Table 1: FormOptiX Technology Stack

Component	Technology	Purpose	Reference
Language	Python 3.11	Core logic	–
ML Clustering	scikit-learn DBSCAN	Floor family detection	[17]
LP Solver	PuLP + CBC Solver	Procurement optimisation	[18]
Data Processing	Pandas, NumPy	Dataframe & array operations	–
Visualisation	Plotly	Interactive charts	–
Frontend	Streamlit	Web browser interface	–

3.2 Time Schedule of Activities – BAR Diagram

The project is planned over an 18-month horizon, divided into Phase 1 (Foundation & Pilot, months 1–9) and Phase 2 (Scale & Integration, months 9–18). A representative 1.5-year Gantt chart is presented in Figure 2.

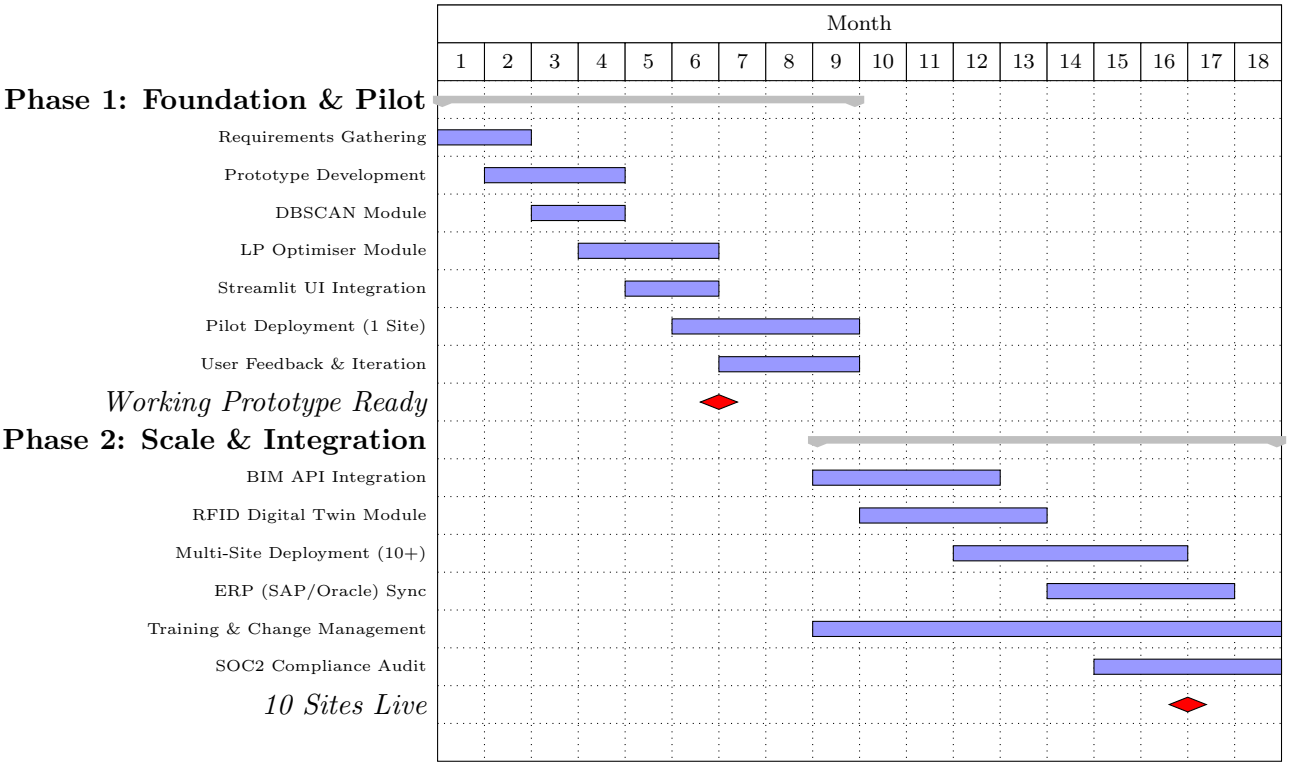

Figure 2: FormOptiX Project BAR (Gantt) Diagram – 18-Month Timeline

Appendix: Competitive Analysis

Table 2: Capability Comparison: FormOptiX vs. Existing Tools

Capability	SAP	Primavera	Doka/PERI	FormOptiX
Repetition Intelligence	×	×	Partial	✓Full
LP BoQ Optimisation	×	×	×	✓Full
Design Freeze Guard	×	×	×	✓Unique
Cross-Site Panel Share	×	×	Manual	✓Auto
Real-Time Analytics	Partial	×	×	✓Full
Cluster-Based Planning	×	×	×	✓Full

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